CENTER FOR DISTANCE AND ONLINE EDUCATION
INTEGRAL UNIVERSITY, LUCKNOW

A Project Report
on

"Hospital Appointment Management System"

Submitted by
Name of Student
Enrollment number: Enrollment Number
Program: MCA
Semester: _____
Session: January 2026

Under the Supervision of
Name of the Guide
Designation and Department

Submitted in partial fulfillment of the requirement for the award of Degree

Declaration

I, Name of Student, hereby declare that the Project Report titled "Hospital Appointment Management System" submitted to Centre for Distance and Online Education, Integral University, Lucknow in partial fulfillment of the requirement for the award of the degree MCA is my original work.

The findings and conclusions presented in this report are based on project work carried out by me. The work has not been submitted elsewhere for any degree, diploma, certificate, or similar academic award. If anything is found incorrect at a later stage, the university reserves the right to take appropriate action as per its rules.

Date: _____
Place: _____

Signature: _____
Name of Student: _____
Enrollment Number: _____

Certificate from Supervisor

This is to certify that the Project Report titled "Hospital Appointment Management System" submitted by Name of Student, Enrollment No. Enrollment Number, is an original work carried out under my direct supervision in partial fulfillment of the requirements for the degree MCA at Centre for Distance and Online Education, Integral University, Lucknow.

To the best of my knowledge, this project report has not been submitted for any other award. The student has demonstrated satisfactory performance during the project period and has completed the system design, implementation, testing, and documentation activities required for the project.

Date: _____

Signature: _____
Name: _____
Qualification: _____
Designation: _____
Institution: _____

Acknowledgement

I express my sincere gratitude to Centre for Distance and Online Education, Integral University, Lucknow for providing the opportunity to prepare this project report on "Hospital Appointment Management System". The work helped me understand the practical application of web technologies, database design, authentication, routing, and role based access control in a real world healthcare scenario.

I am thankful to my supervisor for valuable guidance, suggestions, and encouragement throughout the project. I also acknowledge the support of teachers, classmates, and family members who motivated me during the development and documentation process.

This project strengthened my understanding of full stack development. It also helped me learn how software can solve common administrative problems faced by patients, doctors, and hospital staff.

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Abstract

The Hospital Appointment Management System is developed to provide an online platform where patients can search doctors by department, select available appointment slots, and submit booking information through a simple interface. The system also gives doctors and administrators separate dashboards for managing appointments and schedules.

The project uses React for the user interface, Express for server side APIs, MySQL for relational data storage, and JSON Web Token authentication for secure access. The application includes patient registration, login, doctor listing, department management, appointment booking, appointment cancellation, status updates, schedule management, and dashboard statistics.

The expected outcome of the project is an efficient appointment workflow that reduces manual errors and improves visibility for all participants. Patients get faster access to doctors, doctors can view appointment requests, and administrators can manage departments, doctors, schedules, and patient records from one system.

Keywords: Hospital Management, Appointment Booking, React, Express, MySQL, JWT, Patient Portal, Doctor Dashboard, Admin Dashboard.

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Chapter I: Introduction - Background of the Study

The Hospital Appointment Management System is a web based project designed to simplify the process of finding doctors, checking available appointment slots, booking consultations, and managing hospital administrative records. The application uses a React frontend, an Express backend, JWT based authentication, and a MySQL database. It supports three major roles: administrator, doctor, and patient.

The project focuses on reducing manual work in appointment handling. In a traditional environment, patients often wait in queues, staff members maintain registers, and doctors receive appointment information through informal communication. A computerized appointment system improves transparency, decreases duplication, and makes appointment history easier to maintain.

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This section explains background of the study with reference to the hospital appointment process. The system is designed for institutions where doctors, departments, patients, and appointment slots must be coordinated accurately. The application gives users role based screens and stores all important information in a relational database.

The frontend provides a clear booking flow: select a department, choose a doctor, select a date, check available slots, enter the problem description, and confirm the appointment. If a patient is not logged in, the system collects patient details and creates the patient account before confirming the booking.

The backend exposes REST APIs for authentication, departments, doctors, schedules, appointments, and patients. This separation makes the project modular and easier to test. Each route handles a specific responsibility and communicates with MySQL through parameterized queries.

Chapter I: Introduction - Healthcare and Digital Transformation

The Hospital Appointment Management System is a web based project designed to simplify the process of finding doctors, checking available appointment slots, booking consultations, and managing hospital administrative records. The application uses a React frontend, an Express backend, JWT based authentication, and a MySQL database. It supports three major roles: administrator, doctor, and patient.

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Chapter I: Introduction - Need for Appointment Management

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Chapter I: Introduction - Overview of the Proposed System

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Chapter I: Introduction - Scope of the Project

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Chapter I: Introduction - Users of the System

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Chapter I: Introduction - Technology Context

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Chapter I: Introduction - Literature Review

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Chapter I: Introduction - Existing System Review

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Chapter I: Introduction - Organization of the Report

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Chapter II: Objectives of the Study - Main Objective

The main objective is to design and implement an online appointment management system that allows patients to book doctor appointments and allows hospital staff to manage records efficiently.

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The objectives are specific and measurable. Each module in the system maps to one or more objectives. For example, the appointment module satisfies the booking and tracking objective, while the schedule module supports doctor availability management.

Clear objectives also help during testing. A feature can be considered complete only when it satisfies the related objective from the user perspective and from the data management perspective.

Chapter II: Objectives of the Study - Patient Objectives

Patients should be able to register, login, search departments and doctors, view consultation information, check available time slots, book appointments, cancel appointments, and view appointment history.

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Chapter II: Objectives of the Study - Doctor Objectives

Doctors should be able to login securely, view appointment requests assigned to them, approve or reject appointments, mark appointments as completed, and manage availability through schedules.

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Chapter II: Objectives of the Study - Admin Objectives

The administrator should be able to manage departments, add doctors, view patients, monitor appointment statistics, and maintain the overall operation of the system.

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Chapter II: Objectives of the Study - Technical Objectives

The project should demonstrate full stack development using React, Express, MySQL, JWT authentication, REST APIs, reusable UI components, and responsive design.

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Chapter II: Objectives of the Study - Quality Objectives

The system should be simple, reliable, secure, maintainable, and scalable enough for future enhancements such as payments, notifications, prescriptions, and report exports.

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Chapter III: Problem Statement - Manual Appointment Difficulties

In many hospitals, appointment management still depends on physical registers, telephone calls, and manual coordination. Such processes are slow and error prone. Patients may not know which doctor is available, staff may accidentally allocate the same slot to multiple patients, and doctors may not receive organized daily appointment lists.

The issue of manual appointment difficulties affects the quality of service. A patient centric healthcare environment requires clear information, quick booking, and reliable records. Without a centralized system, appointment status, patient details, and doctor schedules can become scattered across multiple sources.

The proposed system addresses these problems by maintaining departments, doctors, patients, schedules, and appointment records in one database. It also gives each user role a dashboard that displays only the actions relevant to that role.

The problem can be summarized as the need for a secure, online, role based, and database driven appointment management platform for hospital operations.

Chapter III: Problem Statement - Long Queues and Waiting Time

In many hospitals, appointment management still depends on physical registers, telephone calls, and manual coordination. Such processes are slow and error prone. Patients may not know which doctor is available, staff may accidentally allocate the same slot to multiple patients, and doctors may not receive organized daily appointment lists.

The issue of long queues and waiting time affects the quality of service. A patient centric healthcare environment requires clear information, quick booking, and reliable records. Without a centralized system, appointment status, patient details, and doctor schedules can become scattered across multiple sources.

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Chapter III: Problem Statement - Slot Conflict and Duplication

In many hospitals, appointment management still depends on physical registers, telephone calls, and manual coordination. Such processes are slow and error prone. Patients may not know which doctor is available, staff may accidentally allocate the same slot to multiple patients, and doctors may not receive organized daily appointment lists.

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Chapter III: Problem Statement - Limited Access to Doctor Information

In many hospitals, appointment management still depends on physical registers, telephone calls, and manual coordination. Such processes are slow and error prone. Patients may not know which doctor is available, staff may accidentally allocate the same slot to multiple patients, and doctors may not receive organized daily appointment lists.

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The problem can be summarized as the need for a secure, online, role based, and database driven appointment management platform for hospital operations.

Chapter III: Problem Statement - Weak Record Management

In many hospitals, appointment management still depends on physical registers, telephone calls, and manual coordination. Such processes are slow and error prone. Patients may not know which doctor is available, staff may accidentally allocate the same slot to multiple patients, and doctors may not receive organized daily appointment lists.

The issue of weak record management affects the quality of service. A patient centric healthcare environment requires clear information, quick booking, and reliable records. Without a centralized system, appointment status, patient details, and doctor schedules can become scattered across multiple sources.

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The problem can be summarized as the need for a secure, online, role based, and database driven appointment management platform for hospital operations.

Chapter III: Problem Statement - Communication Gaps

In many hospitals, appointment management still depends on physical registers, telephone calls, and manual coordination. Such processes are slow and error prone. Patients may not know which doctor is available, staff may accidentally allocate the same slot to multiple patients, and doctors may not receive organized daily appointment lists.

The issue of communication gaps affects the quality of service. A patient centric healthcare environment requires clear information, quick booking, and reliable records. Without a centralized system, appointment status, patient details, and doctor schedules can become scattered across multiple sources.

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The problem can be summarized as the need for a secure, online, role based, and database driven appointment management platform for hospital operations.

Chapter III: Problem Statement - Problem Summary

In many hospitals, appointment management still depends on physical registers, telephone calls, and manual coordination. Such processes are slow and error prone. Patients may not know which doctor is available, staff may accidentally allocate the same slot to multiple patients, and doctors may not receive organized daily appointment lists.

The issue of problem summary affects the quality of service. A patient centric healthcare environment requires clear information, quick booking, and reliable records. Without a centralized system, appointment status, patient details, and doctor schedules can become scattered across multiple sources.

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The problem can be summarized as the need for a secure, online, role based, and database driven appointment management platform for hospital operations.

Chapter IV: Methodology - Research Design

This section explains the methodology followed for research design in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Requirement Analysis

This section explains the methodology followed for requirement analysis in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Functional Requirements

This section explains the methodology followed for functional requirements in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Non Functional Requirements

This section explains the methodology followed for non functional requirements in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - System Architecture

This section explains the methodology followed for system architecture in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Frontend Design

This section explains the methodology followed for frontend design in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

The frontend is implemented in React with Vite. It maintains state for authentication, departments, doctors, selected doctor, selected appointment date, available slots, appointment records, and dashboard information.

The interface includes public landing and booking, login and registration, dashboards, doctor search, appointment table, department management, doctor management, schedule management, and patient popup for guest booking.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Backend API Design

This section explains the methodology followed for backend api design in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

The backend is implemented with Express. Routes are grouped into auth, departments, doctors, schedules, appointments, and patients. Each route performs database operations and returns JSON responses to the frontend.

JWT authentication protects private routes. Admin routes check for admin role, appointment actions check for patient, doctor, or admin role, and schedule operations are available to doctors and administrators.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Database Design

This section explains the methodology followed for database design in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

The database contains users, departments, doctors, patients, doctor_schedules, appointments, medical_records, and notifications. Foreign keys connect doctors to users and departments, patients to users, and appointments to patients, doctors, and departments.

Important fields include appointment_no, appointment_date, appointment_time, problem_description, status, available_date, start_time, end_time, and slot_duration. These fields make the appointment workflow traceable and searchable.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Authentication Design

This section explains the methodology followed for authentication design in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Role Based Access Control

This section explains the methodology followed for role based access control in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Appointment Booking Flow

This section explains the methodology followed for appointment booking flow in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Schedule Management Flow

This section explains the methodology followed for schedule management flow in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Dashboard Design

This section explains the methodology followed for dashboard design in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Validation and Error Handling

This section explains the methodology followed for validation and error handling in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Testing Methodology

This section explains the methodology followed for testing methodology in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter IV: Methodology - Deployment Methodology

This section explains the methodology followed for deployment methodology in the Hospital Appointment Management System. The project follows a modular full stack approach in which the frontend, backend, and database are developed as separate but connected layers.

The development process started by identifying users and their tasks. The major users are administrator, doctor, patient, and public visitor. Their tasks were converted into modules and then into APIs and screens.

The methodology emphasizes simplicity, maintainability, and correctness. Instead of mixing all responsibilities in one place, the project separates public data loading, private dashboard data, appointment creation, schedule updates, and administrative actions.

This approach makes the application easier to understand and supports future modifications. New features such as online payment or SMS notifications can be added by extending existing modules.

Chapter V: Data Analysis and Interpretation - User Role Analysis

This section analyzes user role analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Department Data Analysis

This section analyzes department data analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Doctor Data Analysis

This section analyzes doctor data analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Patient Data Analysis

This section analyzes patient data analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Appointment Status Analysis

This section analyzes appointment status analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Schedule and Slot Analysis

This section analyzes schedule and slot analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Dashboard Statistics

This section analyzes dashboard statistics for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Booking Flow Analysis

This section analyzes booking flow analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Authentication Analysis

This section analyzes authentication analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Data Integrity Analysis

This section analyzes data integrity analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Security Analysis

This section analyzes security analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Usability Analysis

This section analyzes usability analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Performance Analysis

This section analyzes performance analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Maintainability Analysis

This section analyzes maintainability analysis for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter V: Data Analysis and Interpretation - Summary of Findings

This section analyzes summary of findings for the project. The analysis is based on the implemented modules, database structure, and expected use cases in a hospital appointment environment.

The users table stores login identities and role information. The doctors table connects doctor profiles to departments. The patients table stores demographic information. The appointments table records doctor, patient, department, date, time, problem description, and appointment status.

The appointment status values are pending, approved, rejected, cancelled, and completed. These values make it possible to interpret the progress of a booking from request to final consultation. The dashboard uses status counts to show operational workload.

The system also supports default slots when a custom schedule is not found. This design makes the booking flow usable even if the administrator has not entered a schedule for every date. Custom schedules can override default availability.

Interpretation of the data shows that a centralized appointment system improves accuracy, avoids slot duplication, and gives management a quick view of hospital activity.

Interpretation Table

Module	Purpose
Users	Stores login and role
Doctors	Stores medical profile
Appointments	Stores booking lifecycle
Schedules	Stores doctor availability

Chapter VI: Discussion and Results - Implemented Features

The discussion of implemented features shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VI: Discussion and Results - Role Wise Results

The discussion of role wise results shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VI: Discussion and Results - Patient Booking Result

The discussion of patient booking result shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VI: Discussion and Results - Doctor Dashboard Result

The discussion of doctor dashboard result shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VI: Discussion and Results - Admin Management Result

The discussion of admin management result shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VI: Discussion and Results - System Strengths

The discussion of system strengths shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VI: Discussion and Results - Limitations

The discussion of limitations shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VI: Discussion and Results - Overall Result

The discussion of overall result shows that the project meets the main requirements of an appointment management system. The system provides a structured flow from doctor discovery to appointment confirmation and administrative monitoring.

The public booking flow reduces friction because patients can begin booking without logging in first. Patient details are collected only when the appointment is ready to be confirmed. This improves usability and keeps the first step simple.

Doctors and administrators receive controlled access through JWT based authentication. A doctor can view and update assigned appointments, while an administrator can manage departments, doctors, schedules, and patients. This role separation improves security and clarity.

The result of the project is a working full stack application that can be extended for real hospital use after further validation, hosting, backup planning, notification integration, and user acceptance testing.

Chapter VII: Conclusion - Conclusion Overview

The Hospital Appointment Management System successfully demonstrates how a web based system can improve the appointment process in a healthcare environment. It reduces dependency on manual registers and provides role based access to patients, doctors, and administrators.

The project achieved its objectives by implementing patient registration, login, department and doctor listing, slot checking, appointment booking, appointment cancellation, appointment status update, schedule management, patient listing, and dashboard statistics.

From a technical perspective, the project improved understanding of React state management, REST API design, Express routing, MySQL relational modeling, JWT authentication, and frontend backend integration.

The final conclusion is that the system is a practical and extendable solution for small and medium healthcare institutions that need a reliable digital appointment process.

Chapter VII: Conclusion - Achievement of Objectives

The Hospital Appointment Management System successfully demonstrates how a web based system can improve the appointment process in a healthcare environment. It reduces dependency on manual registers and provides role based access to patients, doctors, and administrators.

The project achieved its objectives by implementing patient registration, login, department and doctor listing, slot checking, appointment booking, appointment cancellation, appointment status update, schedule management, patient listing, and dashboard statistics.

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The final conclusion is that the system is a practical and extendable solution for small and medium healthcare institutions that need a reliable digital appointment process.

Chapter VII: Conclusion - Learning Outcomes

The Hospital Appointment Management System successfully demonstrates how a web based system can improve the appointment process in a healthcare environment. It reduces dependency on manual registers and provides role based access to patients, doctors, and administrators.

The project achieved its objectives by implementing patient registration, login, department and doctor listing, slot checking, appointment booking, appointment cancellation, appointment status update, schedule management, patient listing, and dashboard statistics.

From a technical perspective, the project improved understanding of React state management, REST API design, Express routing, MySQL relational modeling, JWT authentication, and frontend backend integration.

The final conclusion is that the system is a practical and extendable solution for small and medium healthcare institutions that need a reliable digital appointment process.

Chapter VII: Conclusion - Final Conclusion

The Hospital Appointment Management System successfully demonstrates how a web based system can improve the appointment process in a healthcare environment. It reduces dependency on manual registers and provides role based access to patients, doctors, and administrators.

The project achieved its objectives by implementing patient registration, login, department and doctor listing, slot checking, appointment booking, appointment cancellation, appointment status update, schedule management, patient listing, and dashboard statistics.

From a technical perspective, the project improved understanding of React state management, REST API design, Express routing, MySQL relational modeling, JWT authentication, and frontend backend integration.

The final conclusion is that the system is a practical and extendable solution for small and medium healthcare institutions that need a reliable digital appointment process.

Chapter VIII: Recommendations and Suggestions - Recommendations

The system can be improved by adding SMS and email notifications so patients receive booking confirmation, reminder messages, cancellation alerts, and status updates. Notifications would reduce missed appointments and improve communication.

Online payment integration can be added for consultation fees. A payment gateway would allow patients to pay during booking and administrators to track paid and unpaid appointments.

Future versions can include prescription upload, medical record viewing, doctor leave management, reports export, hospital branch management, teleconsultation links, and advanced analytics.

Before deployment in a real hospital, the system should undergo security review, user acceptance testing, database backup planning, hosting configuration, and privacy compliance review.

Chapter VIII: Recommendations and Suggestions - Suggestions for Future Work

The system can be improved by adding SMS and email notifications so patients receive booking confirmation, reminder messages, cancellation alerts, and status updates. Notifications would reduce missed appointments and improve communication.

Online payment integration can be added for consultation fees. A payment gateway would allow patients to pay during booking and administrators to track paid and unpaid appointments.

Future versions can include prescription upload, medical record viewing, doctor leave management, reports export, hospital branch management, teleconsultation links, and advanced analytics.

Before deployment in a real hospital, the system should undergo security review, user acceptance testing, database backup planning, hosting configuration, and privacy compliance review.

Chapter VIII: Recommendations and Suggestions - Implementation Roadmap

The system can be improved by adding SMS and email notifications so patients receive booking confirmation, reminder messages, cancellation alerts, and status updates. Notifications would reduce missed appointments and improve communication.

Online payment integration can be added for consultation fees. A payment gateway would allow patients to pay during booking and administrators to track paid and unpaid appointments.

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Before deployment in a real hospital, the system should undergo security review, user acceptance testing, database backup planning, hosting configuration, and privacy compliance review.

References / Bibliography

React Documentation. Component based user interface development and state management concepts.

Express Documentation. Routing, middleware, request handling, and REST API development.

MySQL Documentation. Relational database design, constraints, joins, and SQL queries.

JSON Web Token Documentation. Token based authentication and authorization patterns.

Vite Documentation. Modern frontend tooling and production build workflow.

Project source code of Hospital Appointment Management System: React frontend, Express backend, and MySQL schema.

Additional references include class notes, project guidelines provided by the university, and standard web development practices followed during implementation.

All references were used for academic learning and project preparation. The final implementation is customized for the Hospital Appointment Management System project.

References / Bibliography Continued

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Credentials / Demo Accounts

This page lists the demo credentials and access information that can be used by an examiner or evaluator to review the Hospital Appointment Management System. The credentials are intended only for academic demonstration and testing.

Application URL: http://localhost:5173. Backend API URL: http://localhost:5000/api. The frontend should be started from the client folder and the backend should be started from the backend folder after configuring the MySQL database and environment variables.

Administrator login: Email admin@hospital.com, Password password123. The administrator can view dashboard statistics, manage departments, register doctors, create doctor schedules, update slot duration, monitor appointments, and review payment amounts.

Doctor login: Email aisha@hospital.com, Password password123. Doctor login: Email rahul@hospital.com, Password password123. A doctor can view assigned appointments, update appointment status, and manage only his or her own schedule.

Patient login: Email priya@example.com, Password password123. A patient can search doctors by department, check available slots for the upcoming five days, enter a problem description, pay the consultation fee, and view booked appointments in the patient dashboard.

Payment testing uses Razorpay test mode. The project allows UPI and card payment options in the checkout interface. Payment records are stored with appointment details so administrators and doctors can see paid amounts in their dashboards.

For printed submission, this credentials page should be included at the end of the report. If real deployment credentials are created later, this page should be replaced with secure demonstration accounts and should never include production secrets.